H112 — second-digit p^3 halving carry

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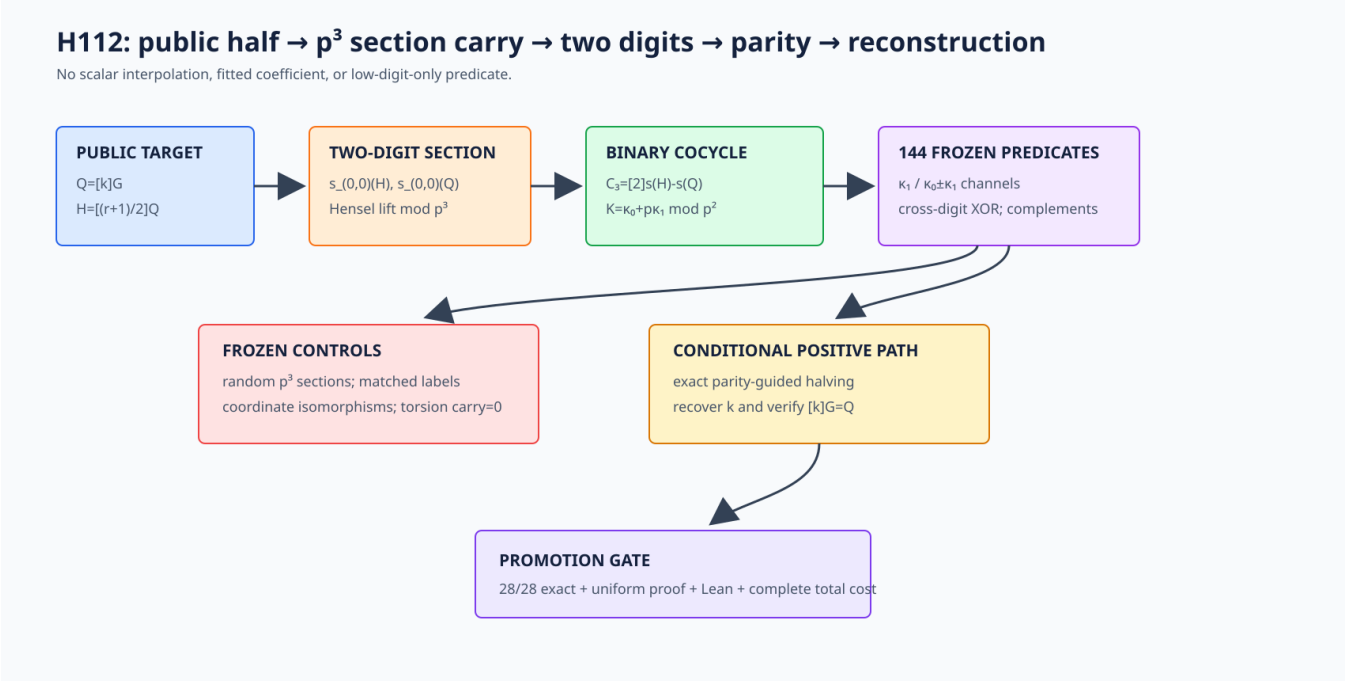


Figure 1: H112 second-digit carry evidence flow

Editable vector source: `h112-p3-second-digit-carry-flow.svg`.

Requirement-to-evidence contract

- **Mechanism-new relative to H111:** lift to `Z/p^3Z` and retain the second formal digit and cross-digit structure. Do not reuse a low-digit-only predicate rejected by H111.
- **Exact candidate:** freeze 144 constant public predicates before execution. No fitted coefficients, curve-sized lookup tables, or scalar interpolation.
- **Measurement:** all seven curves, generator relabelings `1,2,3,5`, every nonzero target, complete scoring, eligible repeated-halving reconstruction, and total costs.
- **Controls:** randomized two-digit sections, symmetry-matched labels, coordinate isomorphisms, exact two-layer torsion-section zero carries, and generic square-root references.
- **Proof:** Lean must cover reduction-kernel membership, digit decomposition, conditional parity/reconstruction, EC verification, and charged cost.
- **PR:** paired draft PRs must contain this freeze before execution.

Mechanism

For `Q=[k]G`, let `H=[(r+1)/2]Q`. Define a two-digit Hensel section by

```
x(s_(u1,u2)(Q)) = x(Q) + p*u1 + p^2*u2  (mod p^3),
y(s_(u1,u2)(Q)) = the unique Hensel lift congruent to y(Q) (mod p).
```

The canonical section is `s_(0,0)`. Its halving carry and normalized formal word are

```
C_3(Q) = [2]s_(0,0)(H) - s_(0,0)(Q),
K(Q)   = t(C_3(Q))/p mod p^2,
K(Q)   = kappa_0(Q) + p*kappa_1(Q),  0 <= kappa_i < p.
```

`kappa_0` is H111's first-layer word. H112 asks whether the genuinely new second digit, or a fixed joint function of both digits, retains the canonical integer wrap distinguishing even `k/2` from odd `(k+r)/2`.

Frozen 144-predicate family

For a field element `z`, define four bits exactly as in H111:

`zero(z), integer_lsb(z), centered(z), legendre(z).`

Each bit is tested with its complement. Freeze two subfamilies:

1. **Depth channels (72):** for each channel `kappa_1`, `kappa_0+kappa_1`, `kappa_1-kappa_0`, each offset `c in {-1,0,1}`, each of four bits, and complement: `3*3*4*2 = 72`.
2. **Cross-digit XOR (72):** for offsets `c,d in {-1,0,1}` and each of four bit kinds, XOR the bit of `kappa_0+c` with the same bit of `kappa_1+d`, then test it and its complement: `3*3*4*2 = 72`.

Total: 144. No predicate depending only on `kappa_0` is included. The target label is canonical scalar parity `k mod 2`.

An exact predicate invokes the same public repeated-halving decoder as H111:

`R <- [(r+1)/2] (R - [b]G).`

After at most `ceil(log2 r)` queries, verify the reconstructed `k` by `[k]G=Q`.

Frozen workload and controls

1. Execute all 28 rows and all 9,096 nonzero targets.
2. Check the `p^3` Hensel equation, lifted doubling, mixed-projective kernel subtraction, projective equation, reduction to identity, unit `Y`, formal divisibility by `p`, and digit recombination.
3. Score all 144 formulas completely and retain all ties and exact predicates.
4. On E101 and E127, run eight deterministic randomized two-digit section controls and eight symmetry-matched parity-label controls each.
5. Repeat primary canonical words after coordinate scaling by `lambda=2`.
6. Solve the unique two-digit prime-to-`p` torsion lifts of `H` and `Q`. The first correction digit must kill `kappa_0`; the second must kill `kappa_1`; their doubled-half carry must be exact identity modulo `p^3`.
7. Run full reconstruction only for exact rows. A primary positive must be stable across all four generator relabelings; a generic gate needs 28/28.

Stopping is exhaustive. Any arithmetic, digit, torsion, control, decoder, or EC mismatch invalidates the run.

Cost and promotion gates

Charge both Hensel digits, lifted group operations modulo `p^3`, formal extraction, predicate work, public EC halvings, setup, controls, memory, advice, and final verification. A uniform exact predicate would give `O(log r)` queries with polynomial bit complexity and hence a subexponential algorithm.

Promotion requires 28/28 exactness, complete reconstruction and EC verification, a uniform proof of the actual predicate, compiled Lean interfaces, complete charged complexity, and a source-complete novelty review.

Prior-art and interpretation boundary

The bounded screen retains Smart's anomalous-curve lifting attack (DOI), Huang–Kueh–Tan (DOI), Silverman's lifting survey (DOI), and Yasuda's formal-group analysis (DOI). They establish the lifting/formal-group context. This bounded screen did not locate the exact second-digit halving-carry test. External novelty is not established.

Failure rejects only these 144 functions of this canonical `p^3` word. A toy positive is not a generic theorem; control-matched fit is not EC-specific evidence; and a zero torsion-section carry is a control, not a formal-log oracle.